

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459954.9772 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.053
Transit depth (Rp/Rs)^2	3.81 +/- 2.05 [%]
Orbital Inclination (inc)	78.85 +/- 2.75
Airmass coefficient 1 (a1)	0.785 +/- 0.032
Airmass coefficient 2 (a2)	0.169 +/- 0.064
Scatter in the residuals of the lightcurve fit is	4.08 %
Best Comparison Star	#3 - [350, 182]
Optimal Aperture	2.16
Optimal Annulus	10.79
Transit Duration (day)	0.036 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.053 vs 0.1594 (deviation 0.67σ)
Duration fit vs published	0.036 d vs 0.0512 d (deviation 0.85σ)
Mid-time O–C	-5.6 min
Depth SNR	3.7
Residual scatter	4.08 %

Light curve

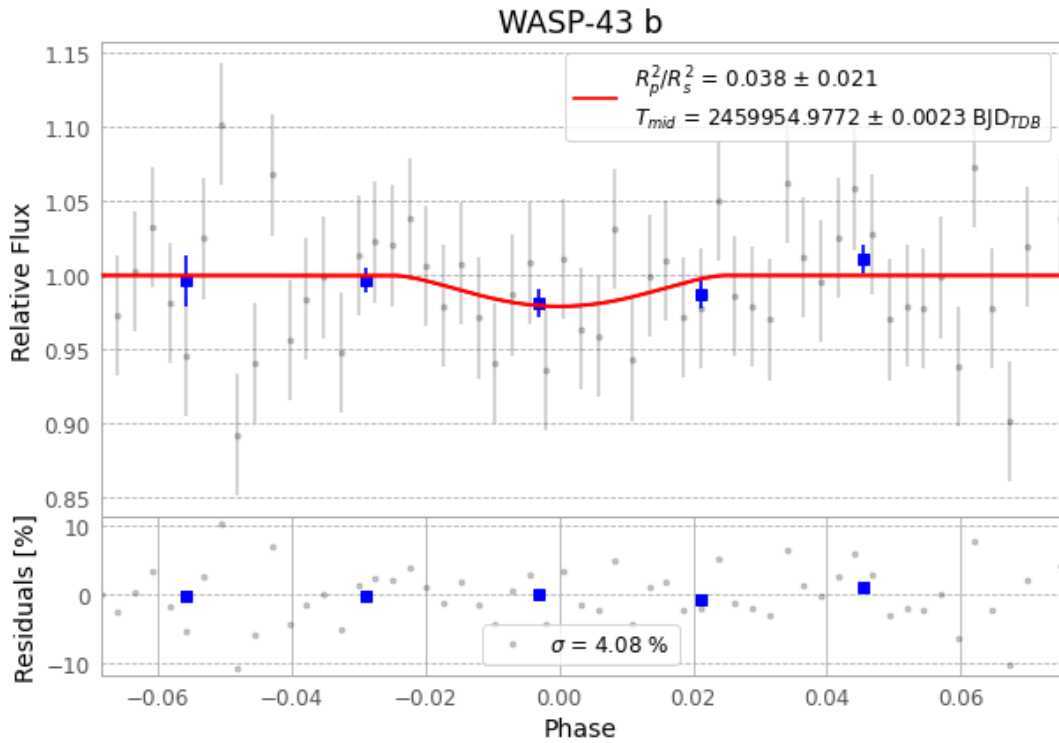


Figure 1 — FinalLightCurve_WASP-43 b_2023-01-10.png (SHA-256 514eca5112930346...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 183], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 748fdb1dd29c8c61...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

62 FITS frames (41 MB), WASP-43230110100015.FITS → WASP-43230110130315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230110.log (e6396edde41c...), FinalLightCurve_WASP-43 b_2023-01-10.png (514eca511293...), FinalLightCurve_WASP-43 b_2023-01-10.csv (ae1927382fe4...)

Compute resources

Wall time 365.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 20:37Z.